

Mini Project- Big Basket Data Analysis

Steps to Follow

Step 1: Load DataSet.

Step 2: Use head function to look for first 12 rows.

Step 3: Get Description of the data in the DataFrame.

Step 4: Find Information about the DataFrame.

Step 5: Find out Top & least sold products.

Step 6: Measuring discount on a certain item.

Step 7: Find out the Missing Values from the Dataset.

Step 8: Find out the outliers from the dataset according to the columns and fill them with the mean.

Step 9: Create Plots or visualizations.

Exploratory Data Analysis (EDA)

Importing Libraries :

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Step 1 : Load DataSet.

```
df=pd.read_csv("D:/BigBasket Products.csv")
df
```

	index	product \
0	1	Garlic Oil - Vegetarian Capsule 500 mg
1	2	Water Bottle - Orange
2	3	Brass Angle Deep - Plain, No.2
3	4	Cereal Flip Lid Container/Storage Jar - Assort...
4	5	Creme Soft Soap - For Hands & Body
...	...	...
27550	27551	Wottagirl! Perfume Spray - Heaven, Classic
27551	27552	Rosemary

27552	27553	Peri-Peri Sweet Potato Chips
27553	27554	Green Tea - Pure Original
27554	27555	United Dreams Go Far Deodorant

	category	sub_category \
0	Beauty & Hygiene	Hair Care
1	Kitchen, Garden & Pets	Storage & Accessories
2	Cleaning & Household	Pooja Needs
3	Cleaning & Household	Bins & Bathroom Ware
4	Beauty & Hygiene	Bath & Hand Wash
...	...	...
27550	Beauty & Hygiene	Fragrances & Deos
27551	Gourmet & World Food	Cooking & Baking Needs
27552	Gourmet & World Food	Snacks, Dry Fruits, Nuts
27553	Beverages	Tea
27554	Beauty & Hygiene	Men's Grooming

	brand	sale_price	market_price	Discount%
\				
0	Sri Sri Ayurveda	220.00	220.0	0.000000
1	Mastercook	180.00	180.0	0.000000
2	Trm	119.00	250.0	52.400000
3	Nakoda	149.00	176.0	15.340909
4	Nivea	162.00	162.0	0.000000
...	...	...	...	...
27550	Layerr	199.20	249.0	20.000000
27551	Puramate	67.50	75.0	10.000000
27552	FabBox	200.00	200.0	0.000000
27553	Tetley	396.00	495.0	20.000000
27554	United Colors Of Benetton	214.53	390.0	44.992308

	type	rating \
0	Hair Oil & Serum	4.1
1	Water & Fridge Bottles	2.3
2	Lamp & Lamp Oil	3.4
3	Laundry, Storage Baskets	3.7
4	Bathing Bars & Soaps	4.4
...	...	...
27550	Perfume	3.9
27551	Herbs, Seasonings & Rubs	4.0

```

27552      Nachos & Chips      3.8
27553      Tea Bags      4.2
27554      Men's Deodorants      4.5

```

description

```

0      This Product contains Garlic Oil that is known...
1      Each product is microwave safe (without lid), ...
2      A perfect gift for all occasions, be it your m...
3      Multipurpose container with an attractive desi...
4      Nivea Creme Soft Soap gives your skin the best...
...
27550      Layerr brings you Wottagirl Classic fragrant b...
27551      Puramate rosemary is enough to transform a dis...
27552      We have taken the richness of Sweet Potatoes (...
27553      Tetley Green Tea with its refreshing pure, ori...
27554      The new mens fragrance from the United Dreams ...

```

[27555 rows x 11 columns]

Step 2: Use head function to look for first 12 rows.

```
df.head(12)
```

```

      index      product \
0         1  Garlic Oil - Vegetarian Capsule 500 mg
1         2      Water Bottle - Orange
2         3  Brass Angle Deep - Plain, No.2
3         4  Cereal Flip Lid Container/Storage Jar - Assort...
4         5  Creme Soft Soap - For Hands & Body
5         6  Germ - Removal Multipurpose Wipes
6         7      Multani Mati
7         8  Hand Sanitizer - 70% Alcohol Base
8         9  Biotin & Collagen Volumizing Hair Shampoo + Bi...
9        10  Scrub Pad - Anti- Bacterial, Regular
10        11  Wheat Grass Powder - Raw
11        12  Butter Cookies Gold Collection

```

	category	sub_category	brand
0	Beauty & Hygiene	Hair Care	Sri Sri Ayurveda
1	Kitchen, Garden & Pets	Storage & Accessories	Mastercook
2	Cleaning & Household	Pooja Needs	Trm
3	Cleaning & Household	Bins & Bathroom Ware	Nakoda
4	Beauty & Hygiene	Bath & Hand Wash	Nivea
5	Cleaning & Household	All Purpose Cleaners	Nature Protect

6	Beauty & Hygiene	Skin Care	Satinance
7	Beauty & Hygiene	Bath & Hand Wash	Bionova
8	Beauty & Hygiene	Hair Care	StBotanica
9	Cleaning & Household	Mops, Brushes & Scrubs	Scotch brite
10	Gourmet & World Food	Cooking & Baking Needs	NUTRASHIL
11	Gourmet & World Food	Chocolates & Biscuits	Sapphire
	sale_price	market_price	Discount%
\			
0	220.0	220.0	0.000000
1	180.0	180.0	0.000000
2	119.0	250.0	52.400000
3	149.0	176.0	15.340909
4	162.0	162.0	0.000000
5	169.0	199.0	15.075377
6	58.0	58.0	0.000000
7	250.0	250.0	0.000000
8	1098.0	1098.0	0.000000
9	20.0	20.0	0.000000
10	261.0	290.0	10.000000
11	600.0	600.0	0.000000
	rating	description	
0	4.1	This Product contains Garlic Oil that is known...	
1	2.3	Each product is microwave safe (without lid), ...	
2	3.4	A perfect gift for all occasions, be it your m...	
3	3.7	Multipurpose container with an attractive desi...	
4	4.4	Nivea Creme Soft Soap gives your skin the best...	
5	3.3	Stay protected from contamination with Multipu...	
6	3.6	Satinance multani matti is an excellent skin t...	
7	4.0	70%Alcohol based is gentle of hand leaves skin...	
8	3.5	An exclusive blend with Vitamin B7 Biotin, Hyd...	

```

9      4.3  Scotch Brite Anti- Bacterial Scrub Pad thoroug...
10     4.0  Wheatgrass is a superfood potent health food w...
11     2.2  Enjoy a tin full of delicious butter cookies m...

```

Step 3: Get Description of the data in the DataFrame.

```
df.describe()
```

	index	sale_price	market_price	Discount%
rating				
count	27555.000000	27549.000000	27555.000000	27555.000000
18919.000000				
mean	13778.000000	334.648391	382.056664	8.641330
3.943295				
std	7954.58767	1202.102113	581.730717	420.566982
0.739217				
min	1.000000	2.450000	3.000000	-68813.953490
1.000000				
25%	6889.500000	95.000000	100.000000	0.000000
3.700000				
50%	13778.000000	190.320000	220.000000	5.000000
4.100000				
75%	20666.500000	359.000000	425.000000	20.000000
4.300000				
max	27555.000000	112475.000000	12500.000000	100.000000
5.000000				

Step 4: Find information about the DataFrame

```
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27555 entries, 0 to 27554
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  -
0   index                 27555 non-null  int64
1   product               27554 non-null  object
2   category              27555 non-null  object
3   sub_category          27555 non-null  object
4   brand                 27554 non-null  object
5   sale_price            27549 non-null  float64
6   market_price          27555 non-null  float64
7   Discount%             27555 non-null  float64
8   type                  27555 non-null  object
9   rating                18919 non-null  float64
10  description            27440 non-null  object
dtypes: float64(4), int64(1), object(6)
memory usage: 2.3+ MB

```

Step 5: a) Find out Top sold products.

```
df["product"].value_counts().head(3)
```

```
product
Turmeric Powder/Arisina Pudi      26
Extra Virgin Olive Oil            15
Cow Ghee/Tuppa                   14
Name: count, dtype: int64
```

b) Find out least sold products.

```
df["product"].value_counts().tail(3)
```

```
product
Disney Mickey Mouse Plastic Kids Sipper Bottle - Coolest Dude      1
Chocolates-Roasted Peanut Chocolate                                1
Green Tea - Pure Original                                           1
Name: count, dtype: int64
```

Step 6: Measuring discount on a certain item

```
df["Discount"]=((df["market_price"]-df["sale_price"])/
df["market_price"]*100).round(2)
df.head(5)
```

	index	product \
0	1	Garlic Oil - Vegetarian Capsule 500 mg
1	2	Water Bottle - Orange
2	3	Brass Angle Deep - Plain, No.2
3	4	Cereal Flip Lid Container/Storage Jar - Assort...
4	5	Creame Soft Soap - For Hands & Body

	category	sub_category	brand \
0	Beauty & Hygiene	Hair Care	Sri Sri Ayurveda
1	Kitchen, Garden & Pets	Storage & Accessories	Mastercook
2	Cleaning & Household	Pooja Needs	Trm
3	Cleaning & Household	Bins & Bathroom Ware	Nakoda
4	Beauty & Hygiene	Bath & Hand Wash	Nivea

	sale_price	market_price	Discount%	type
rating \				
0	220.0	220.0	0.000000	Hair Oil & Serum
4.1				
1	180.0	180.0	0.000000	Water & Fridge Bottles
2.3				
2	119.0	250.0	52.400000	Lamp & Lamp Oil
3.4				
3	149.0	176.0	15.340909	Laundry, Storage Baskets
3.7				

4	162.0	162.0	0.000000	Bathing Bars & Soaps
---	-------	-------	----------	----------------------

4.4

	description	Discount
0	This Product contains Garlic Oil that is known...	0.00
1	Each product is microwave safe (without lid), ...	0.00
2	A perfect gift for all occasions, be it your m...	52.40
3	Multipurpose container with an attractive desi...	15.34
4	Nivea Creme Soft Soap gives your skin the best...	0.00

Step 7: Find out the missing values of the dataset

```
missing_values=df.isnull().sum()
missing_values
```

```
index          0
product        1
category       0
sub_category   0
brand          1
sale_price     6
market_price   0
Discount%      0
type           0
rating        8636
description    115
Discount       6
dtype: int64
```

```
df[df['product'].isnull()]
```

	index	product	category	sub_category	brand
sale_price \					
14363	14364	NaN	Beverages	Coffee	Cothas Coffee
200.0					

	market_price	Discount%	type	rating \
14363	240.0	16.666667	Ground Coffee	4.2

	description	Discount
14363	Cothas Specialty Blend Coffee and Chicory incl...	16.67

```
df[df['brand'].isnull()]
```

	index	product	category \
9765	9766	Food Package - Medium	Cleaning & Household

	sub_category	brand	sale_price	market_price
Discount% \				
9765	Disposables, Garbage Bag	NaN	50.0	50.0

0.0

		type	rating	description	Discount
9765	Aluminium Foil, Clingwrap		NaN	NaN	0.0

```
df[df['sale_price'].isnull()]
```

	index	product \
1719	1720	Puja Flower Wicks - Puvvu Vathulu Batti
1720	1721	Powder - Sambar
1721	1722	Steel Fork - Medium, Premium Excel Series, BBCL08
1722	1723	Snack Mix - Dhokla
2401	2402	Battery AA 3UT Hi Top
2402	2403	Klassic Plain Cocktail Napkins (22 X 22 cm)

	category	sub_category
brand \		
1719	Cleaning & Household	Pooja Needs Jaji
1720	Foodgrains, Oil & Masala	Masalas & Spices Annapoorna
1721	Kitchen, Garden & Pets	Crockery & Cutlery BB Home
1722	Snacks & Branded Foods	Ready To Cook & Eat MTR
2401	Kitchen, Garden & Pets	Appliances & Electricals Nippo
2402	Cleaning & Household	Disposables, Garbage Bag Origami

	sale_price	market_price	Discount%	type
rating \				
1719	NaN	20.0	100.0	Camphor & Wicks
NaN				
1720	NaN	46.0	100.0	Blended Masalas
NaN				
1721	NaN	299.0	100.0	Cutlery, Spoon & Fork
NaN				
1722	NaN	65.0	100.0	Breakfast & Snack Mixes
4.1				
2401	NaN	150.0	100.0	Battery & Electrical
4.0				
2402	NaN	32.0	100.0	Toilet Paper
3.9				

	description	Discount
1719	Jaji Puja Flower Batti / Puvvu Vathulu is made...	NaN
1720	Annaporna Spices are the preserve for Connoiss...	NaN
1721	BB Home provides fine and classy cutlery that ...	NaN
1722	MTR Dhokla is extremely special in Gujarat. It...	NaN

2401	This Nippo battery incorporates state-of-the-a...	NaN
2402	Klassic Plain Cocktail Napkins (22 X 22 cm) 10...	NaN

```
df[df['rating'].isnull()]
```

	index	product \
55	56	Soothing Cucumber Facial Scrub With Apricot Seeds
59	60	Corporate Planner Diary With Premium PU Leathe...
65	66	Ayurvedic Anti-Tan Face Pack
68	69	Organic Carom Seeds/Ajwain/0m Kalu
69	70	Padded Harness - 3/4 inch, Grey Colour
...	...	...
27509	27510	Deluxe Crackers - Veg
27511	27512	Specialist Stain Remover Pen & Marker
27514	27515	Verge & Sheer Perfume For Pair
27530	27531	Tick'et to Fleadom Dry Shampoo For Dogs
27531	27532	Vanilla Ice Cream

	category	sub_category	
brand \			
55	Beauty & Hygiene	Skin Care	
TJ0RI			
59	Cleaning & Household	Stationery	Prozo
Plus			
65	Beauty & Hygiene	Skin Care	
TJ0RI			
68	Foodgrains, Oil & Masala	Masalas & Spices	
Earthon			
69	Kitchen, Garden & Pets	Pet Food & Accessories	
Glenand			
...	...	...	..
.			
27509	Gourmet & World Food	Chocolates & Biscuits	
Kerk			
27511	Cleaning & Household	All Purpose Cleaners	
365			
27514	Beauty & Hygiene	Fragrances & Deos	Skin by
Titan			
27530	Kitchen, Garden & Pets	Pet Food & Accessories	Captain
Zack			
27531	Gourmet & World Food	Dairy & Cheese	Haagen-Dazs

	sale_price	market_price	Discount%	type
rating \				
55	299.4	499.0	40.0	Face Care
NaN				
59	399.0	399.0	0.0	Notebooks, Files, Folders
NaN				
65	269.4	449.0	40.0	Face Care

NaN				
68	72.0	72.0	0.0	Whole Spices
NaN				
69	840.0	840.0	0.0	Pet Collars & Leashes
NaN				
...	...	...	...	...
...				
27509	150.0	150.0	0.0	Cookies, Biscotti, Wafer
NaN				
27511	449.0	449.0	0.0	Imported Cleaners
NaN				
27514	1615.5	1795.0	10.0	Perfume
NaN				
27530	99.0	99.0	0.0	Pet Cleaning & Grooming
NaN				
27531	225.0	225.0	0.0	Gourmet Ice Cream
NaN				

	description	Discount
55	The soothing feel of cucumber meets the gentle...	40.0
59	A5 Size (210x150mm) \n192 Pages Premium Natura...	0.0
65	A nourishing face pack that removes tan and br...	40.0
68	Earthon's Ajwain is Best quality, organically ...	0.0
69	These are soft padded harness for your active ...	0.0
...	...	...
27509	Kerk Biscuits-has been a household name synony...	0.0
27511	Mightier than the pen. The pen may be mightier...	0.0
27514	VERGE for men paints a picture of a classy out...	10.0
27530	1) No Rinse Defence Against Ticks and Fleas: C...	0.0
27531	Some things are so good in their simplest form...	0.0

[8636 rows x 12 columns]

```
df[df['description'].isnull()]
```

	index	product \
288	289	Arrabbiata Tomato Pasta Sauce With Chilli
480	481	Pumice Stone & Emery Foot Filer - FFL3407, Col...
780	781	Peanut Butter - Chunky
845	846	Squash - Sarasaparilla/Nannari/Sugandhi Root
1047	1048	Natura Rice & Walnut Drink, Vegan
...	...	...
25510	25511	BBPopular Almond/Badam Californian 500gm + BBR...
25583	25584	Chai/Chai/Tea/ Coffee Bone China Microwave Ser...
26116	26117	Tea Strainer - Stainless Steel, Medium
26835	26836	Cold Pressed Virgin Groundnut Oil
27303	27304	Zipper Storage - Gallon

brand \	category	sub_category
---------	----------	--------------

288	Gourmet & World Food	Sauces, Spreads & Dips	
Montanini			
480	Beauty & Hygiene	Bath & Hand Wash	
Kaiv			
780	Gourmet & World Food	Sauces, Spreads & Dips	American
Garden			
845	Beverages	Fruit Juices & Drinks	
NaturoBell			
1047	Gourmet & World Food	Drinks & Beverages	
BORGES			
...	...	...	
...			
25510	Foodgrains, Oil & Masala	Dry Fruits	bb
Combo			
25583	Kitchen, Garden & Pets	Crockery & Cutlery	BP
Bharat			
26116	Kitchen, Garden & Pets	Kitchen Accessories	
Ritu			
26835	Foodgrains, Oil & Masala	Edible Oils & Ghee	
Chekkko			
27303	Cleaning & Household	Disposables, Garbage Bag	
Glad			

	sale_price	market_price	Discount%	
type \				
288	22325.00	325.0	-6769.230769	Mustard & Cheese
Sauces				
480	175.00	175.0	0.000000	Bathing
Accessories				
780	299.00	299.0	0.000000	Chocolate, Peanut
Spread				
845	140.00	140.0	0.000000	Syrups &
Concentrates				
1047	275.01	349.0	21.200573	Health
Drinks				
...	...	...	...	..
.				
25510	443.00	725.0	38.896552	
Almonds				
25583	269.00	373.0	27.882038	Cups, Mugs &
Tumblers				
26116	99.00	135.0	26.666667	Strainer, Ladle,
Spatula				
26835	325.00	325.0	0.000000	Groundnut
Oil				
27303	350.00	350.0	0.000000	Aluminium Foil,
Clingwrap				

rating description Discount

288	5.0	NaN	-6769.23
480	3.8	NaN	0.00
780	4.1	NaN	0.00
845	2.5	NaN	0.00
1047	4.5	NaN	21.20
...	...	...	...
25510	NaN	NaN	38.90
25583	2.5	NaN	27.88
26116	4.3	NaN	26.67
26835	4.2	NaN	0.00
27303	4.7	NaN	0.00

[115 rows x 12 columns]

```
df[df['Discount'].isnull()]
```

	index	product \
1719	1720	Puja Flower Wicks - Puvvu Vathulu Batti
1720	1721	Powder - Sambar
1721	1722	Steel Fork - Medium, Premium Excel Series, BBCL08
1722	1723	Snack Mix - Dhokla
2401	2402	Battery AA 3UT Hi Top
2402	2403	Klassic Plain Cocktail Napkins (22 X 22 cm)

	brand \	category	sub_category
1719		Cleaning & Household	Pooja Needs Jaji
1720		Foodgrains, Oil & Masala	Masalas & Spices Annapoorna
1721		Kitchen, Garden & Pets	Crockery & Cutlery BB Home
1722		Snacks & Branded Foods	Ready To Cook & Eat MTR
2401		Kitchen, Garden & Pets	Appliances & Electricals Nippo
2402		Cleaning & Household	Disposables, Garbage Bag Origami

	sale_price	market_price	Discount%	type
rating \				
1719	NaN	20.0	100.0	Camphor & Wicks
NaN				
1720	NaN	46.0	100.0	Blended Masalas
NaN				
1721	NaN	299.0	100.0	Cutlery, Spoon & Fork
NaN				
1722	NaN	65.0	100.0	Breakfast & Snack Mixes
4.1				
2401	NaN	150.0	100.0	Battery & Electrical
4.0				

2402	NaN	32.0	100.0	Toilet Paper
3.9				
			description	Discount
1719	Jaji Puja Flower Batti / Puvvu Vathulu is made...			NaN
1720	Annaporna Spices are the preserve for Connoiss...			NaN
1721	BB Home provides fine and classy cutlery that ...			NaN
1722	MTR Dhokla is extremely special in Gujarat. It...			NaN
2401	This Nippo battery incorporates state-of-the-a...			NaN
2402	Klassic Plain Cocktail Napkins (22 X 22 cm) 10...			NaN

Find out the Percentage of Missing Values from the Dataset (optional)

```
total_missing_values=missing_values.sum()
total_cell=np.prod(df.shape)
percent_missing = (total_missing_values/total_cell) * 100
print("Percentage: {:.2f}%".format(percent_missing))
```

Percentage: 2.65%

Cleaning the missing values

```
df.loc[df['product'].isna(), 'product']='Unknown'
df['product']

0          Garlic Oil - Vegetarian Capsule 500 mg
1          Water Bottle - Orange
2          Brass Angle Deep - Plain, No.2
3    Cereal Flip Lid Container/Storage Jar - Assort...
4          Creme Soft Soap - For Hands & Body
...
27550    Wottagirl! Perfume Spray - Heaven, Classic
27551          Rosemary
27552    Peri-Peri Sweet Potato Chips
27553    Green Tea - Pure Original
27554    United Dreams Go Far Deodorant
Name: product, Length: 27555, dtype: object

df.loc[df['brand'].isna(), 'brand']='Unknown'
df['brand']

0          Sri Sri Ayurveda
1          Mastercook
2          Trm
3          Nakoda
4          Nivea
...
27550          Layerr
27551    Puramate
27552    FabBox
```

```

27553          Tetley
27554    United Colors Of Benetton
Name: brand, Length: 27555, dtype: object

df.loc[df['sale_price'].isna(), 'sale_price'] = df['sale_price'].mean()
df['sale_price']

0          220.00
1          180.00
2          119.00
3          149.00
4          162.00
...
27550        199.20
27551         67.50
27552        200.00
27553        396.00
27554        214.53
Name: sale_price, Length: 27555, dtype: float64

df.loc[df['rating'].isna(), 'rating'] = df['rating'].mean()
df['rating']

0          4.1
1          2.3
2          3.4
3          3.7
4          4.4
...
27550        3.9
27551        4.0
27552        3.8
27553        4.2
27554        4.5
Name: rating, Length: 27555, dtype: float64

df.loc[df['description'].isna(), 'description'] = 'Unknown'
df['description']

0    This Product contains Garlic Oil that is known...
1    Each product is microwave safe (without lid), ...
2    A perfect gift for all occasions, be it your m...
3    Multipurpose container with an attractive desi...
4    Nivea Creme Soft Soap gives your skin the best...
...
27550    Layerr brings you Wottagirl Classic fragrant b...
27551    Puramate rosemary is enough to transform a dis...
27552    We have taken the richness of Sweet Potatoes (...
27553    Tetley Green Tea with its refreshing pure, ori...
27554    The new mens fragrance from the United Dreams ...
Name: description, Length: 27555, dtype: object

```

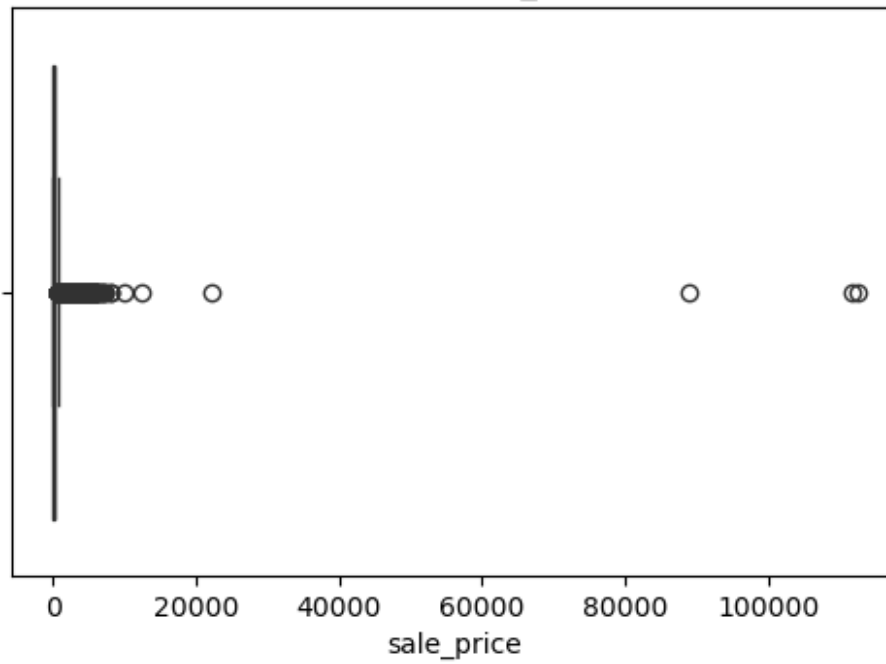
```
missing_values=df.isnull().sum()
missing_values

index      0
product    0
category   0
sub_category 0
brand      0
sale_price 0
market_price 0
Discount%   0
type       0
rating     0
description 0
Discount    6
dtype: int64
```

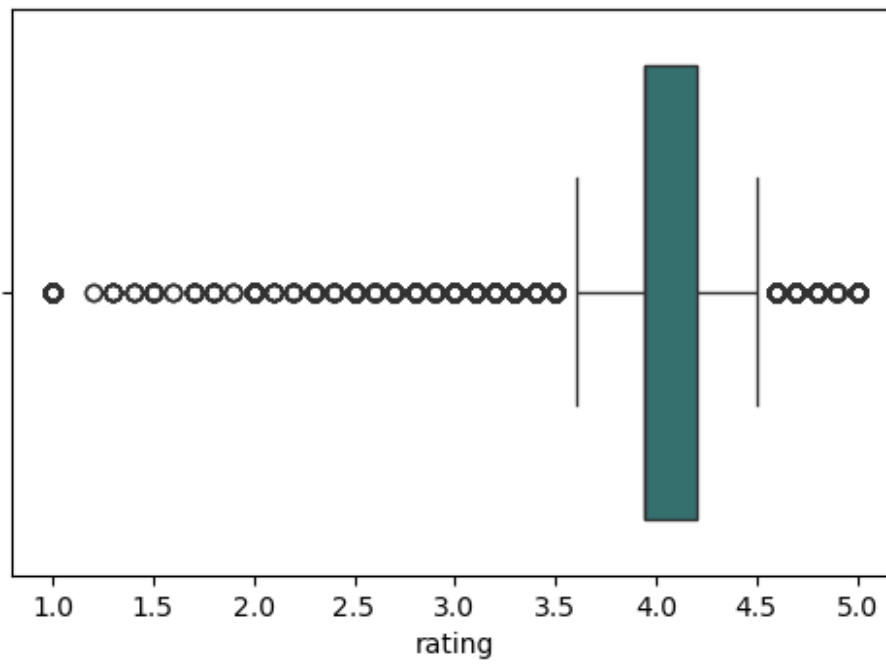
Step 8 : Find out the outliers in the dataset and fill them with mean value

```
numerical_columns = ['sale_price', 'rating', 'market_price', 'Discount %']
plt.figure(figsize=(5, 15))
for i, column in enumerate(numerical_columns, 1):
    plt.subplot(len(numerical_columns), 1, i)
    sns.boxplot(df[column], orient='h', color='#2B7A78')
    plt.title(f'Box Plot - {column}')
    plt.xlabel(column)
    plt.tight_layout()
```

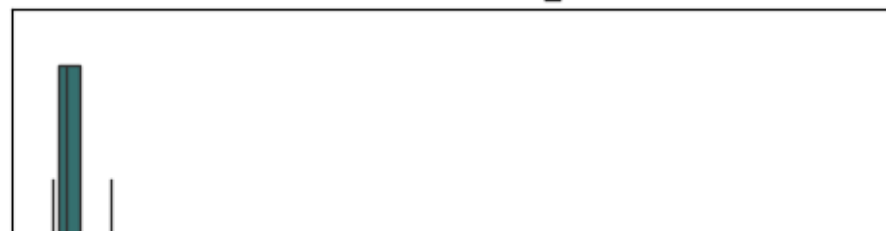
Box Plot - sale_price



Box Plot - rating



Box Plot - market_price



a) Find out the outliers in the dataset

```
def find_outliers(d,col):
    q1=d[col].quantile(0.25)
    q3=d[col].quantile(0.75)
    IQR=q3-q1
    lower_limit= q1-1.5*IQR
    upper_limit= q3+1.5*IQR
    return lower_limit,upper_limit

a= ['sale_price',      'market_price', 'Discount', 'rating']
for i in a:
    print(find_outliers(df,i))

(-264.4725868452575, 694.1209780754292)
(-323.08499673380516, 805.1416612230087)
(-30.0, 50.0)
(3.903924376880428, 4.008912972302903)
```

b) Replacing outliers with mean value

```
df["sale_price"]=df["sale_price"].apply(lambda x:
df["sale_price"].mean() if(x<-267 or x>694.12) else x)
df["sale_price"].head(10)

0      220.000000
1      180.000000
2      119.000000
3      149.000000
4      162.000000
5      169.000000
6       58.000000
7      250.000000
8      334.648391
9       20.000000
Name: sale_price, dtype: float64

df["market_price"]=df["market_price"].apply(lambda x:
df["market_price"].mean() if(x<-323 or x>805) else x)
df["market_price"].tail(10)

27545      349.0
27546      160.0
27547       38.0
27548      499.0
27549       61.0
27550      249.0
27551       75.0
27552      200.0
```

```

27553    495.0
27554    390.0
Name: market_price, dtype: float64

df["Discount"]=df["Discount"].apply(lambda x: df["Discount"].mean()
if(x<-30 or x>50) else x)
df["Discount"].tail(10)

27545    45.85
27546     5.00
27547     0.00
27548     0.00
27549     0.00
27550    20.00
27551    10.00
27552     0.00
27553    20.00
27554    44.99
Name: Discount, dtype: float64

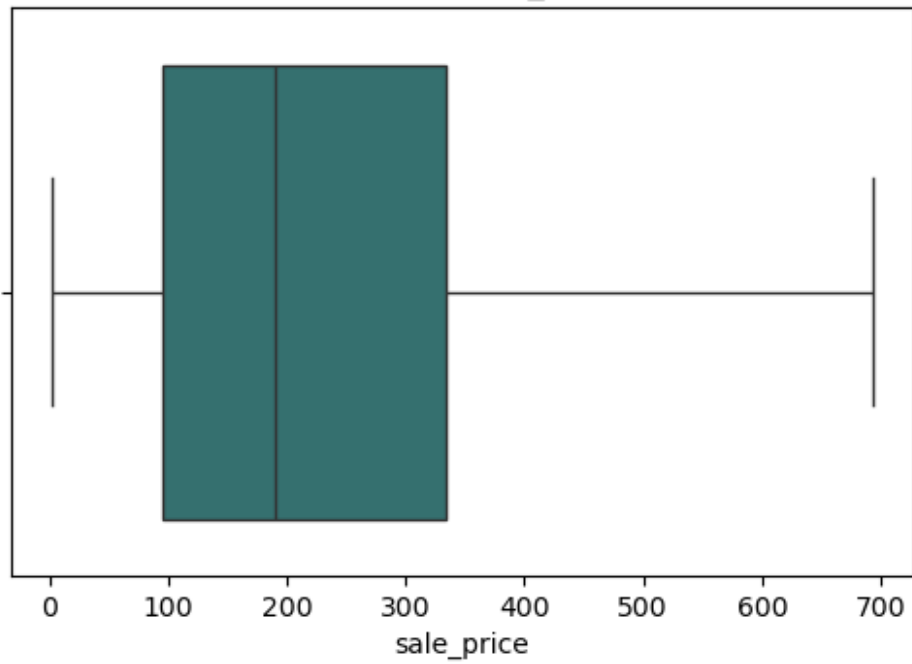
df["rating"]=df["rating"].apply(lambda x: df["rating"].mean()
if(x<3.91 or x>4.00) else x)
df["rating"].head(10)

0    3.969542
1    3.943295
2    3.943295
3    3.969542
4    3.969542
5    3.943295
6    3.969542
7    4.000000
8    3.969542
9    3.969542
Name: rating, dtype: float64

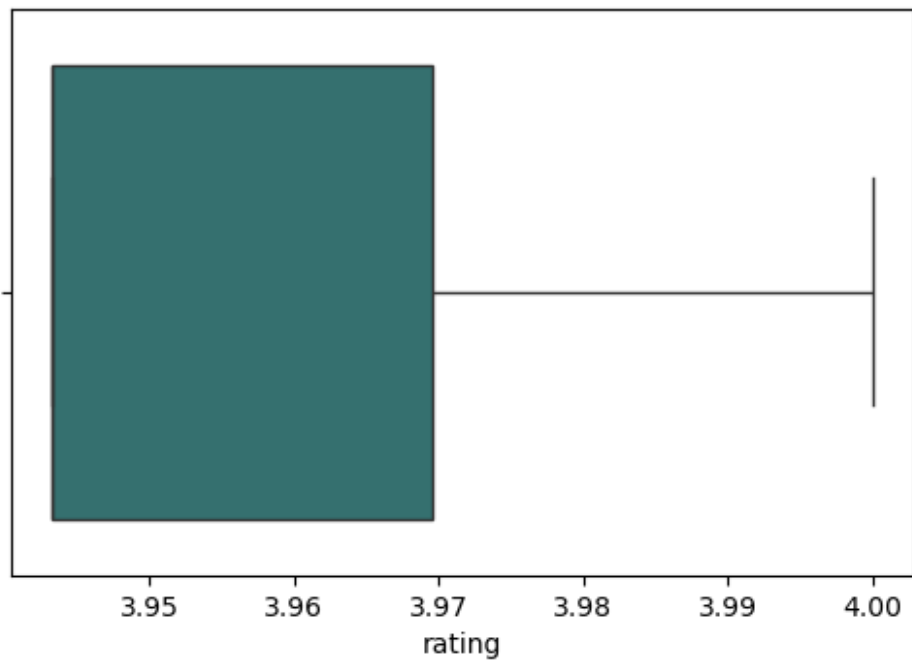
numerical_columns = ['sale_price', 'rating', 'market_price',
'Discount']
# Create box plots for each numerical column
plt.figure(figsize=(5, 15))
for i, column in enumerate(numerical_columns, 1):
    plt.subplot(len(numerical_columns), 1, i) # Create subplots for each
column
    sns.boxplot(df[column], orient='h', color='#2B7A78')
    plt.title(f'Box Plot - {column}')
    plt.xlabel(column)
    plt.tight_layout()

```

Box Plot - sale_price



Box Plot - rating



Box Plot - market_price



Step 9: Create Plots or visualizations.

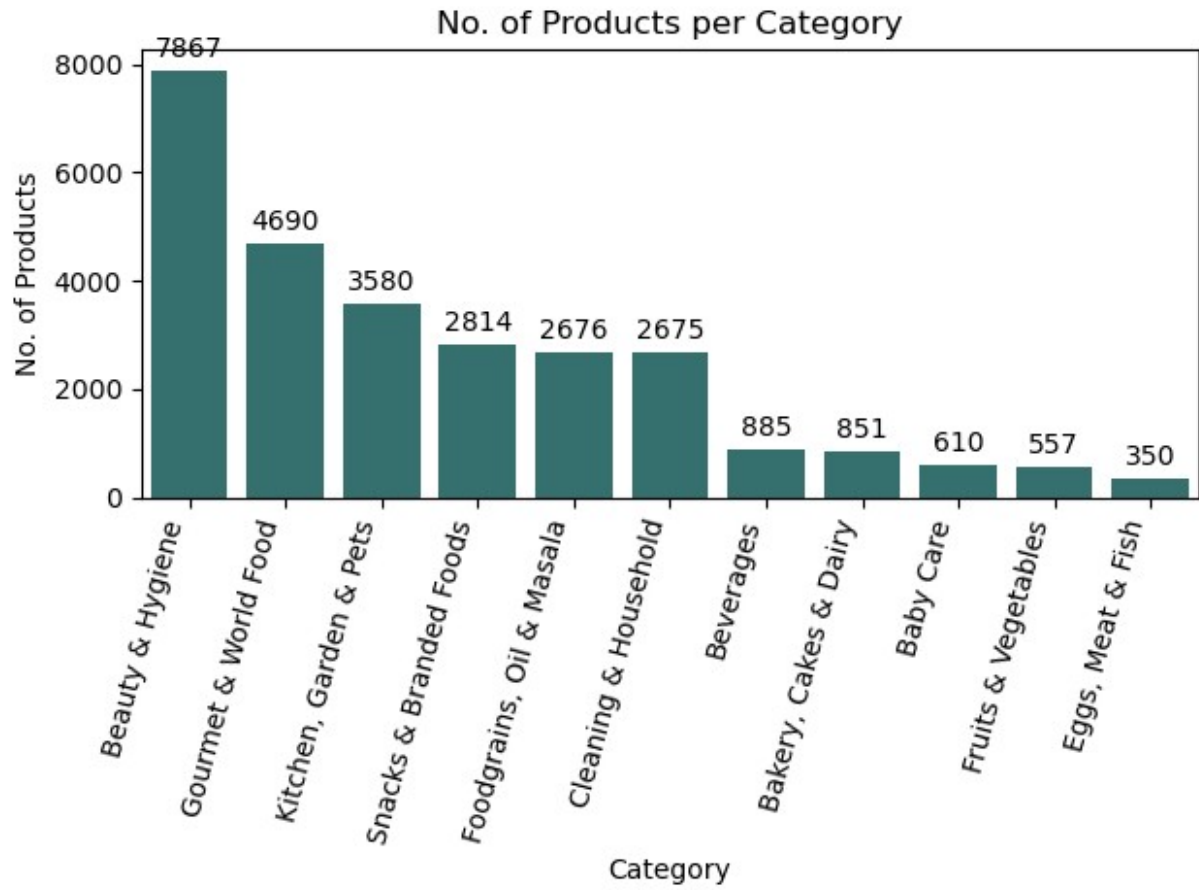
9.1 Number of Products per Category

This bar plot shows the count of products within each category, providing insight into the distribution of products across various categories.

```
counts = df['category'].value_counts()
counts

category
Beauty & Hygiene          7867
Gourmet & World Food      4690
Kitchen, Garden & Pets    3580
Snacks & Branded Foods    2814
Foodgrains, Oil & Masala  2676
Cleaning & Household      2675
Beverages                 885
Bakery, Cakes & Dairy     851
Baby Care                 610
Fruits & Vegetables       557
Eggs, Meat & Fish         350
Name: count, dtype: int64

counts_df = pd.DataFrame({'Category': counts.index, 'Counts':
counts.values})
ax = sns.barplot(x='Category', y='Counts', data=counts_df,
color='#2B7A78')
plt.xticks(rotation=75, ha='right')
ax.set_xlabel('Category')
ax.set_ylabel('No. of Products')
ax.set_title('No. of Products per Category')
for p in ax.patches:
    ax.annotate(f'{int(p.get_height())}', (p.get_x() + p.get_width() /
2., p.get_height()), ha='center', va='baseline',
                fontsize=10, color='black', xytext=(0, 5),
textcoords='offset points')
plt.tight_layout()
plt.show()
```



Key Takeaways:

- Beauty & Hygiene dominates with the highest number of products with a volume of 7876 units.
- Gourmet & World Food and Kitchen, Garden & Pets are average selling products with 4690 and 3580 units, respectively.
- Snacks & Branded Foods have a volume of 2814 Foodgrains followed closely by Oil & Masala, Cleaning & Household which have a similar volume, around 2600 units.
- Beverages, Bakery, Cakes & Dairy are the products with a moderate volume of approximately 900 units.
- Baby Care Fruits & Vegetables have significantly fewer products of around 600 units each.
- Eggs, Meat & Fish has the lowest product count, with 350 units.

Recommendations :

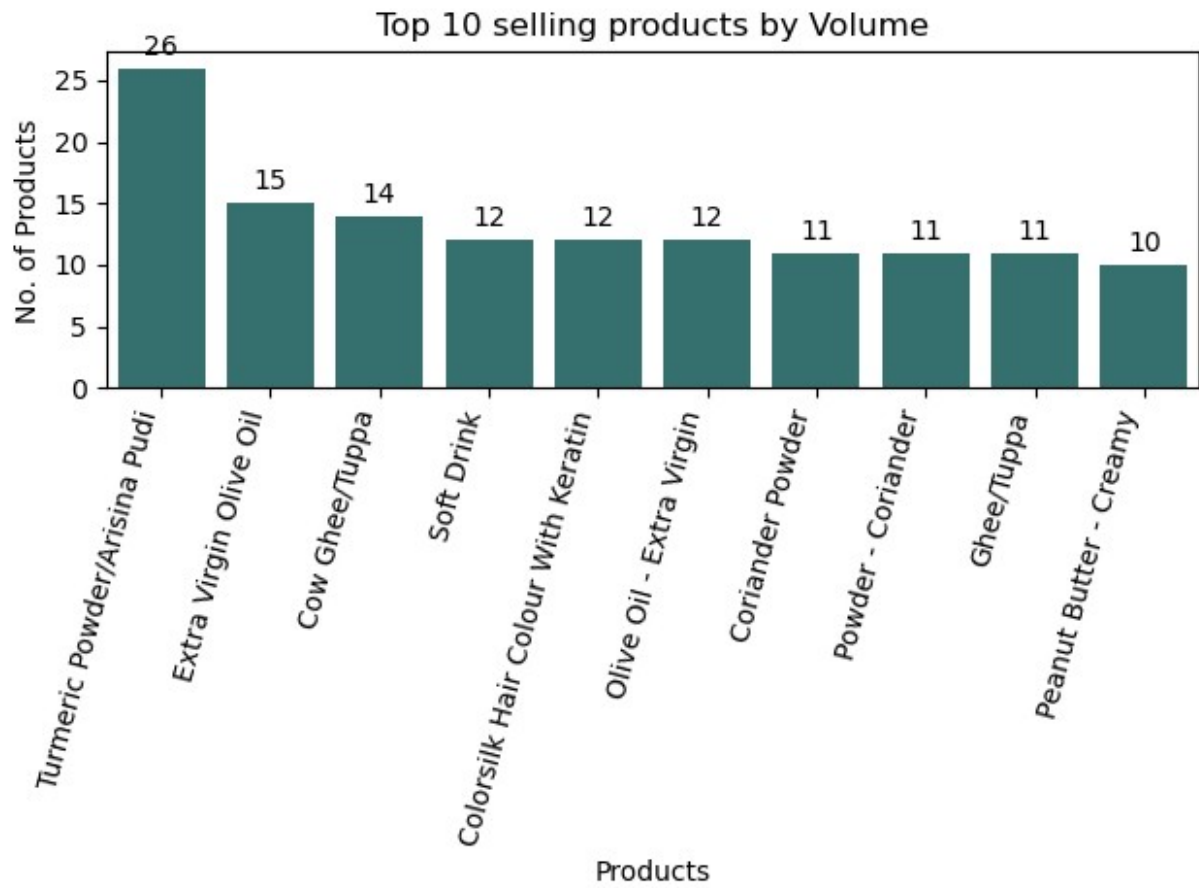
- Focus on high-Performing Products: Continue to promote and invest in Beauty & Hygiene, Gourmet & World Food and Kitchen, Garden & Pets to drive their sales momentum.
- Optimize low-Performing Products: Analyze the performance of products with lower volumes (Eggs, Meat & Fish, Baby Care Fruits & Vegetables, Beverages, Bakery, Cakes & Dairy) to identify opportunities for improvement, such as pricing adjustments or targeted promotions.
- Consider Product Expansion: Explore opportunities to expand the product range in categories with high-performing products to attract a wider customer base.

9.2 Top 10 Selling Products by Volume

This bar plot displays the top 10 products with the highest sales volume, indicating the most popular products among customers

```
counts = df['product'].value_counts()
counts_p = pd.DataFrame({'product': counts.index, 'Counts':
counts.values})[:10]

bx = sns.barplot(x='product', y='Counts', data=counts_p,
color='#2B7A78')
plt.xticks(rotation=75, ha='right')
bx.set_xlabel('Products')
bx.set_ylabel('No. of Products')
bx.set_title('Top 10 selling products by Volume')
for p in bx.patches:
    bx.annotate(f'{int(p.get_height())}', (p.get_x() + p.get_width() /
2., p.get_height()),
ha='center', va='baseline', fontsize=10, color='black', xytext=(0, 5),
textcoords='offset points')
plt.tight_layout()
plt.show()
```



Key Takeaways:

Turmeric Powder/Arisina Pudi is the top-selling product with a significantly higher volume of 26 units.

Extra Virgin Olive Oil is the second-highest selling product with 15 units, followed closely by Cow Ghee/Tuppa with 14 units.

Soft Drink, Olive Oil - Extra Virgin, and Colorsilk Hair Colour With Keratin share the same volume (12 units).

Coriander Powder, Powder - Coriander, and Ghee/Tuppa each have 11 units sold, indicating steady demand.

Casting Creme Gloss Hair Color rounds out the top 10 with 10 units sold, slightly lower than the rest.

Recommendations:

Leverage Top-Selling Products:

Increase promotions and visibility for Turmeric Powder/Arisina Pudi and Extra Virgin Olive Oil to further capitalize on their popularity.

Optimize Moderate-Performing Products:

Analyze factors driving the performance of products like Soft Drink and Colorsilk Hair Colour With Keratin to replicate success in similar categories.

Enhance Sales for Lower-Performing Products:

Investigate the lower sales volume of Casting Creme Gloss Hair Color and develop strategies to increase its appeal, such as offering discounts or showcasing customer testimonials.

9.3 Top 5 Selling Brands by Volume

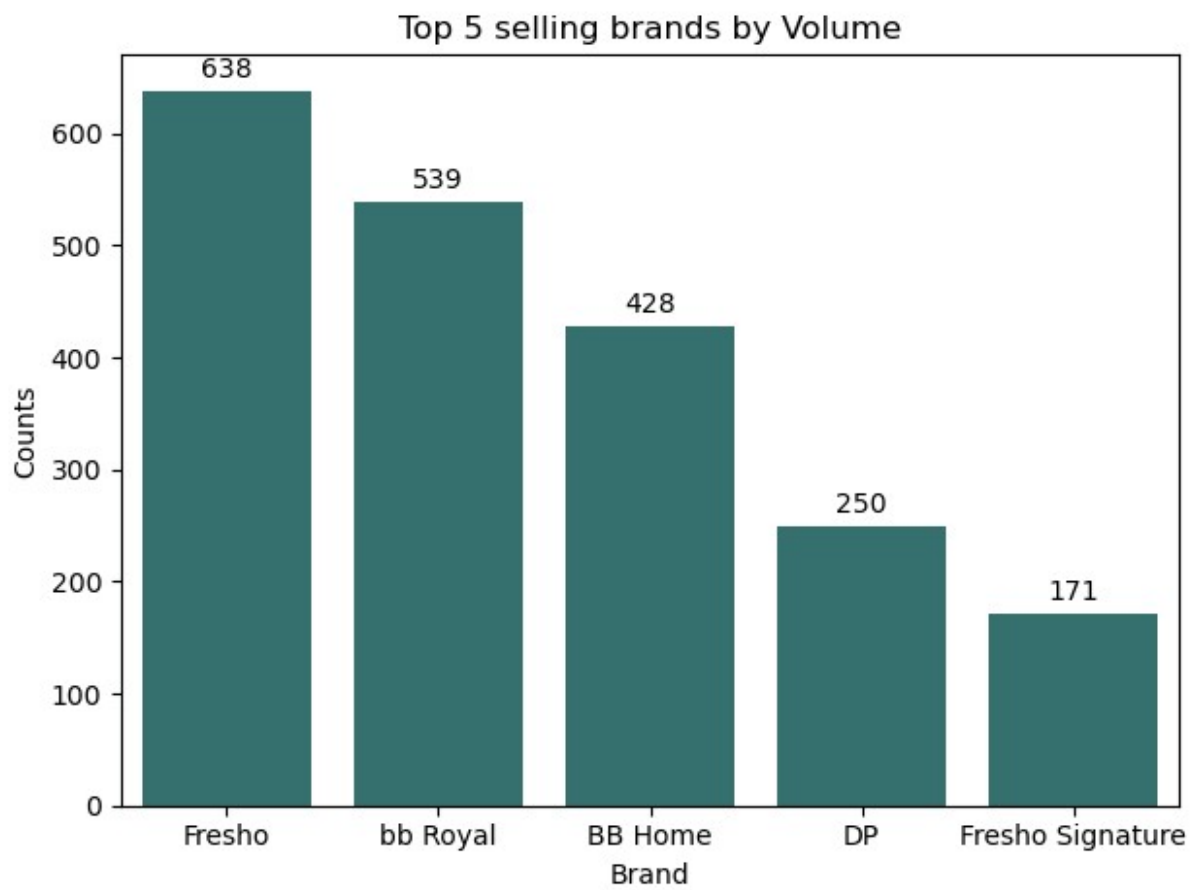
This bar plot illustrates the five brands with the highest sales volume, highlighting the most popular brands.

```
counts = df['brand'].value_counts()
counts_b = pd.DataFrame({'brand': counts.index, 'Counts':
counts.values})[:5]

cx = sns.barplot(x='brand', y='Counts', data=counts_b,
color='#2B7A78')
cx.set_xlabel('Brand')
cx.set_title('Top 5 selling brands by Volume')
for p in cx.patches:
    cx.annotate(f'{int(p.get_height())}', (p.get_x() + p.get_width() /
2., p.get_height()),
ha='center', va='baseline', fontsize=10,
color='black', xytext=(0, 5),
textcoords='offset points')
```



```
plt.tight_layout()  
plt.show()
```



Key Takeaways:

- Fresho is the top-selling brand, with a volume of 638 units.
- bb Royal follows closely with 539 units.
- BB Home Brand is a moderate selling brand with a volume of 428 units.
- DP has a significantly low volume of around 220 units.
- Fresho Signature has the lowest volume among the top 5 brands, with 138 units.

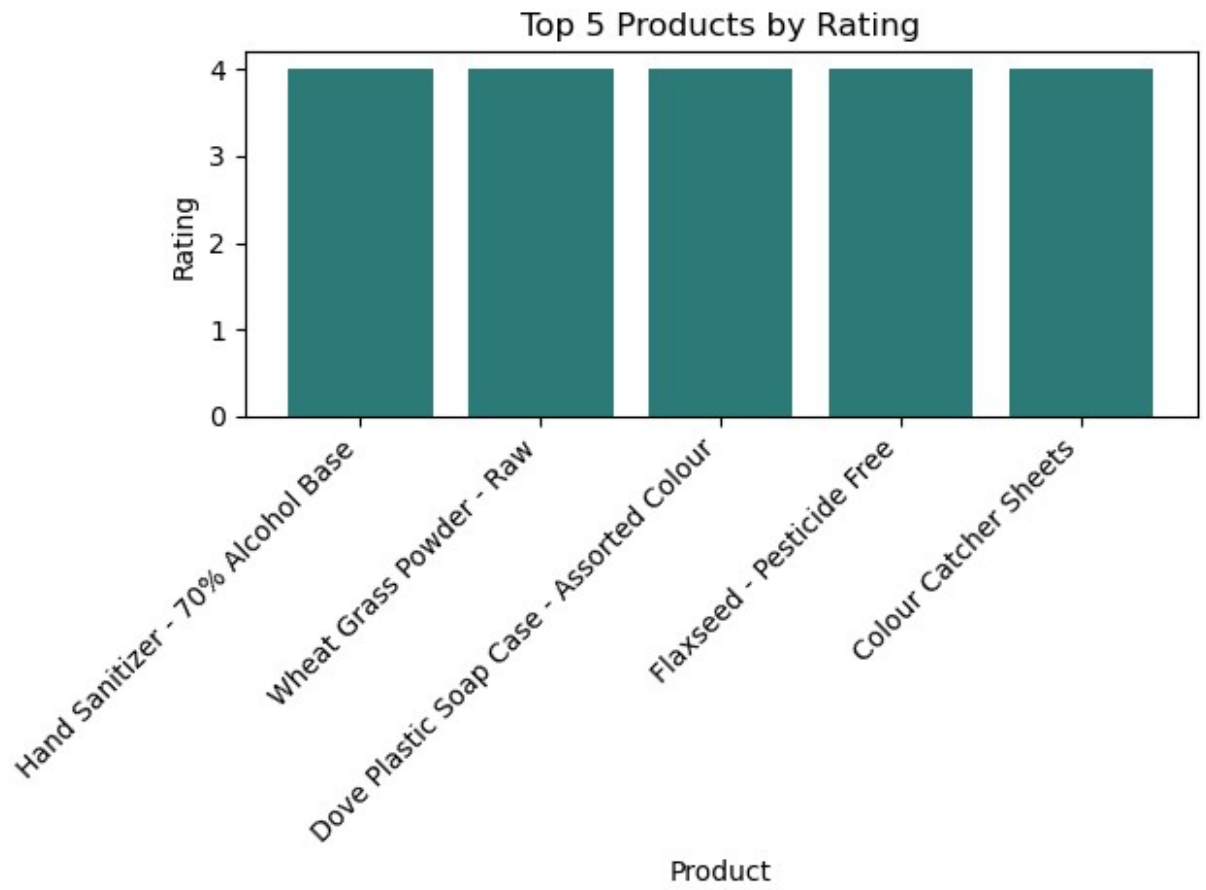
Recommendations :

- Focus on Top-selling brands: Continue to promote and invest in Fresho and bb Royal to maintain their sales momentum.
- Optimize Product Mix: Analyze the performance of products with lower volumes (Fresho Signature, DP) to identify opportunities for improvement, such as pricing adjustments or targeted promotions.
- Consider Product Expansion: Explore opportunities to expand the product range in categories with high-performing products to attract a wider customer base.

9.4 Top 5 Products by Rating:

A bar plot of the highest-rated products, showing the top 5 products with the best ratings

```
top_5_products = df.nlargest(5, 'rating')[['product', 'rating']]
plt.bar(top_5_products['product'], top_5_products['rating'],
color='#2B7A78')
plt.xlabel('Product')
plt.ylabel('Rating')
plt.title('Top 5 Products by Rating')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



Key Takeaways:

Top 5 Products by Rating: All five products have achieved a high rating of 4+, indicating consistent customer satisfaction:

1. Face Wash - Oil Control, Active
2. Smooth Skin Oil - For Dry Skin
3. Veggie Cutter
4. Plain Green Olives
5. Topp Up Milk-Elaichi

Recommendations:

Maintain High Standards: Continue to focus on quality and performance for these products to sustain high customer ratings.

Expand Related Offerings: Introduce variations or complementary products in the same categories (e.g., more types of skincare products or kitchen tools) to capture broader interest.

Leverage Marketing: Offer limited-time discounts or loyalty rewards for these products to boost sales further.

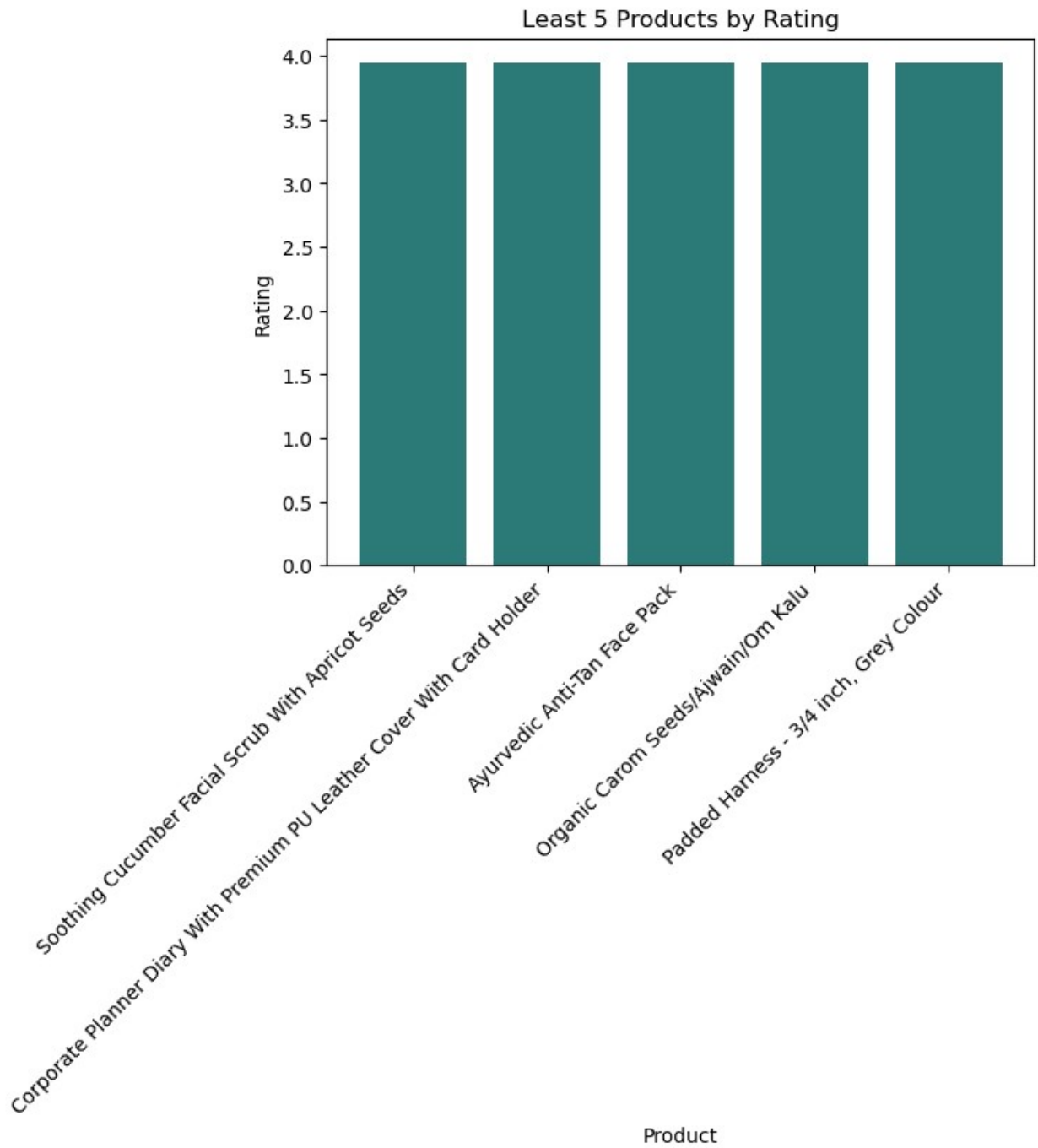
9.5 Least 5 Products by Rating

A bar plot of the least-rated products, showing the top 5 products with the least ratings

```
top_5_products = df.nsmallest(5, 'rating')[['product', 'rating']]
top_5_products
```

	product	rating
55	Soothing Cucumber Facial Scrub With Apricot Seeds	3.943295
59	Corporate Planner Diary With Premium PU Leather...	3.943295
65	Ayurvedic Anti-Tan Face Pack	3.943295
68	Organic Carom Seeds/Ajwain/0m Kalu	3.943295
69	Padded Harness - 3/4 inch, Grey Colour	3.943295

```
top_5_products = df.nsmallest(5, 'rating')[['product', 'rating']]
plt.bar(top_5_products['product'], top_5_products['rating'],
color='#2B7A78')
plt.xlabel('Product')
plt.ylabel('Rating')
plt.title('Least 5 Products by Rating')
plt.xticks(rotation=45, ha='right')
plt.show()
```



Key Takeaways:

All of the following products are the lowest-rated:

1. Soothing Cucumber Facial Scrub With Apricot Seeds
2. Corporate Planner Diary With Premium PU Leather
3. Ayurvedic Anti-Tan Face Pack
4. Organic Carom Seeds/Ajwain/Om Kalu
5. Padded Harness - 3/4 inch, Grey Colour

Recommendations:

Review and Improve Low-Rated Products:

Analyze the feedback for these products to identify areas for improvement, such as quality, packaging, or product description.

Consider Product Removal:

If the issues with low-rated products cannot be resolved, consider removing them from the inventory.

Gather Customer Feedback:

Collect feedback from customers to identify areas for improvement and maintain high ratings.

9.6 Top 10 products with highest discount percentage

A line plot showing the products with the largest discounts, helping identify the items with the highest markdowns.

```
top_10_discounts = df.sort_values(by='Discount%',  
ascending=False).head(10)  
top_10_discounts
```

	index	product \
1722	1723	Snack Mix - Dhokla
1719	1720	Puja Flower Wicks - Puvvu Vathulu Batti
1720	1721	Powder - Sambar
2401	2402	Battery AA 3UT Hi Top
2402	2403	Klassic Plain Cocktail Napkins (22 X 22 cm)
1721	1722	Steel Fork - Medium, Premium Excel Series, BBCL08
26976	26977	Curry Leaves
17713	17714	Fruit & Vegetables Hand Juicer
13318	13319	Small Silicone Spatula With Plastic Handle - A...
13740	13741	Decorative Party Light Big Star String LED Lig...
		category
		sub_category
		brand

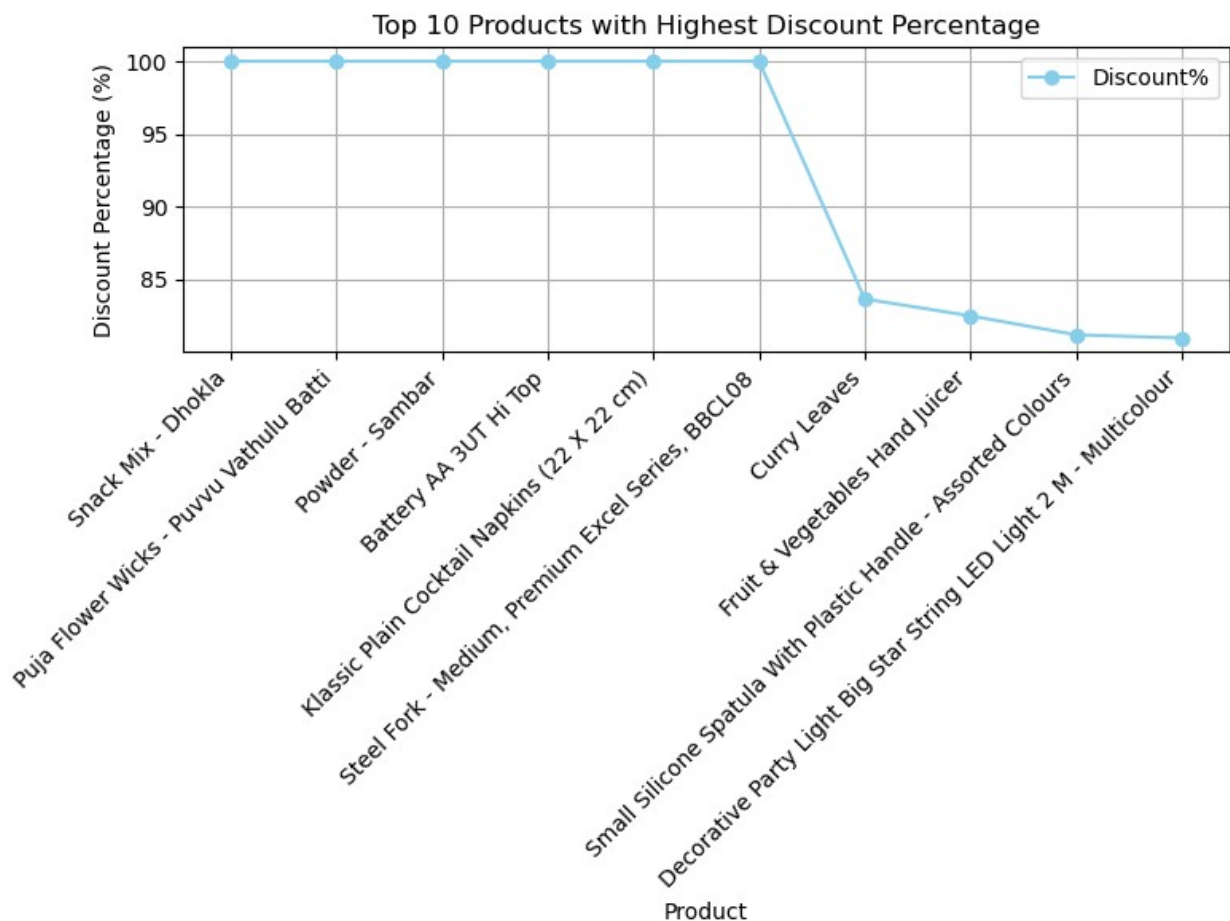
\	1722	Snacks & Branded Foods	Ready To Cook & Eat	MTR
	1719	Cleaning & Household	Pooja Needs	Jaji
	1720	Foodgrains, Oil & Masala	Masalas & Spices	Annapoorna
	2401	Kitchen, Garden & Pets	Appliances & Electricals	Nippo
	2402	Cleaning & Household	Disposables, Garbage Bag	Origami
	1721	Kitchen, Garden & Pets	Crockery & Cutlery	BB Home
	26976	Fruits & Vegetables	Herbs & Seasonings	Fresho
	17713	Kitchen, Garden & Pets	Kitchen Accessories	Triones
	13318	Kitchen, Garden & Pets	Bakeware	HAZEL
	13740	Kitchen, Garden & Pets	Appliances & Electricals	Lexton
		sale_price	market_price	Discount%
	1722	334.648391	65.000000	100.000000
	1719	334.648391	20.000000	100.000000
	1720	334.648391	46.000000	100.000000
	2401	334.648391	150.000000	100.000000
	2402	334.648391	32.000000	100.000000
	1721	334.648391	299.000000	100.000000
	26976	2.450000	15.000000	83.666667
	17713	349.000000	382.056664	82.506266
	13318	75.000000	399.000000	81.203008
	13740	209.000000	382.056664	80.982712
			type	rating
	1722		Breakfast & Snack Mixes	3.969542
	1719		Camphor & Wicks	3.943295
	1720		Blended Masalas	3.943295
	2401		Battery & Electrical	4.000000
	2402		Toilet Paper	3.955550
	1721		Cutlery, Spoon & Fork	3.943295
	26976		Indian & Exotic Herbs	3.943295
	17713		Kitchen Tools & Other Accessories	3.943295
	13318		Baking Tools & Brushes	3.955550
	13740		CFL & Led Bulbs	3.943295
			description	Discount
	1722		MTR Dhokla is extremely special in Gujarat. It...	NaN
	1719		Jaji Puja Flower Batti / Puvvu Vathulu is made...	NaN
	1720		Annaporna Spices are the preserve for Connoiss...	NaN

2401	This Nippo battery incorporates state-of-the-a...	NaN
2402	Klassic Plain Cocktail Napkins (22 X 22 cm) 10...	NaN
1721	BB Home provides fine and classy cutlery that ...	NaN
26976	With dark green and glossy appearance, curry l...	8.621432
17713	Hand juicer for fruits and vegetables in steel...	8.621432
13318	This silicone spatula is safety-tested, heat r...	8.621432
13740	Bring home these bright and festive lights fro...	8.621432

```

top_10_discounts = df.sort_values(by='Discount%',
ascending=False).head(10)
plt.figure(figsize=(8, 6))
plt.plot(top_10_discounts['product'], top_10_discounts['Discount%'],
marker='o', color='skyblue', linestyle='-', label='Discount%')
plt.xticks(rotation=45, ha='right')
plt.xlabel('Product')
plt.ylabel('Discount Percentage (%)')
plt.title('Top 10 Products with Highest Discount Percentage')
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()

```



Key Takeaways:

- Snack Mix - Dhokla, Puja Flower Wicks - Puvvu Vathulu Batti, Powder - Sambar, Battery AA 3UT Hi Top, Klassic Plain Cocktail Napkins (22 X 22 cm) and Steel Fork - Medium, Premium Excel Series, BBCL08 have a discount percentage of 100%.
- Curry Leaves, Fruit & Vegetables Hand Juicer, Small Silicone Spatula With Plastic Handle have almost similar discounts of 82%.
- Decorative Party Light Pine String LED Light has the lowest discount percentage among the top 10 products, at 81%.

Recommendations:

- Leverage High-Discount Products: Promote products with the highest discount percentages to attract customers and boost sales.
- Optimize Discounts: Analyze the impact of discounts on different product categories and adjust discount strategies accordingly to maximize profitability.
- Monitor Competitor Pricing: Stay informed about competitor pricing and adjust your discount strategy to remain competitive.

Overall Takeaways

1. Product Categories

- Beauty & Hygiene: Dominates with the highest product count, followed by Kitchen, Garden & Pets.
- Beverages and Baby Care: Have fewer products, suggesting room for potential expansion.

2. Top-Selling Products

- Turmeric Powder and Cow Ghee are among the top-selling items, showing strong customer demand for essential food products.
- Recommendation: Focus promotions on high-demand products like these to maximize sales.

3. Popular Brands

- bb Royal and BB Home: Leading in sales volume, reflecting customer loyalty toward Big Basket's in-house brands.
- Amul and Himalaya: Perform well, indicating strong market trust.

4. Discounted Products

- Highest Discounts: Products like the Fruit & Vegetables Hand Juicer have discounts up to 82%.
- Recommendation: Leverage these high-discount items to attract price-sensitive customers.

5. Product Ratings

- Highly Rated Products: Face Wash and Veggie Cutter receive top ratings, showing high customer satisfaction.

- Recommendation: Promote these products to boost Big Basket's brand image.

- Low-Rated Products: Items such as Aqua Halo Rejuvenating Conditioner received low ratings, suggesting potential quality issues.

- Recommendation: Address quality concerns in low-rated products to meet customer expectations.

Overall Conclusion

Big Basket's strategy effectively balances high-frequency essentials, strategic discounting, and in-house brand development to build customer loyalty.

Improvements could include:

Addressing quality issues for lower-rated products and expanding the product mix in underperforming categories.